

Loukas Kastanis

MATERIALS SCIENCE & AUTOMATION ENGINEER · PHD, NTUA

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PROFILE

Materials Science & Engineering graduate with a postgraduate specialization in automation systems and PhD research in electronic/electrical materials science. Hands-on experience designing, developing and implementing automation systems for industrial installations — from electrical control panels and PLC/HMI to CNC, CAD/CAM and additive manufacturing.

PROFESSIONAL EXPERIENCE

Mechanical Engineer — CNC & Advanced Manufacturing

ATC Tsampas Bros S.A.

06/2026 – Present

CNC and advanced manufacturing, CAD/CAM, laser and EDM technologies.

Production Engineer

HED Engineering, Aspropyrgos

04/2026 – 06/2026

Designed hooklift-type containers; on-site production support.

Mechanical & Automation Engineer — Industrial Equipment

RAM Europe / Greenmedia, Orchomenos

05/2023 – 04/2026

Design & development of industrial recycling equipment (trommel, conveyors); PLC programming, inverter setup, industrial panel support; field troubleshooting & commissioning. Case studies: robotic sorting system optimization (Yaskawa + hyperspectral vision), industrial press fault diagnosis, trommel design.

Lab Assistant

National Technical University of Athens (NTUA)

09/2014 – 06/2021

Laboratory assistant alongside doctoral research.

Sales Staff

Cosmote e-Value (OTE), Attica

01/2019 – 04/2019

PhD Student — Erasmus+ programme

Nottingham Trent University, United Kingdom

09/2015 – 06/2016

Research on non-volatile memory devices with laser-fabricated nanocrystals.

Assistant Engineer

Petrogaz, Attica

12/2014 – 06/2015

Maintenance and industrial safety, LPG company.

Trainee Engineer

Hellenic Aerospace Industry SA, Schimatari

07/2008 – 09/2008

Non-destructive testing (FPI, MPI, RT, UT, ET), heat treatments, ARC welding, metal depositions, hardness testing, plasma spray, atomic emission spectroscopy, machining, autoclave.

Trainee Engineer

Halyvourgiki, Elefsina

08/2007 - 09/2007

Rebar welding, quality control, XRF.

EDUCATION

PhD in Engineering

National Technical University of Athens (NTUA)

2014 - 2021

MOS Structure with Laser Annealed Metal Nanocrystals Non-Volatile Memory. Doctoral fellowship from the Special Account for Research Funding (ELKE) of NTUA.

MSc, Automation Systems

National Technical University of Athens (NTUA)

2010 - 2011

Simulation of Superconducting Magnetic Shielding using Finite Elements. CAD, CAM, CAE, CNC, automatic control systems.

BSc, Materials Science & Engineering

University of Ioannina

2004 - 2009

Grade 9.13 (First Class) — top of graduating class. Thesis: Galvanic Corrosion and Mechanical Resistance in Aircraft Bonded Repair. Spectroscopy, XRF/XRD, PVD/PLD/CVD, corrosion & fatigue testing.

SKILLS & TOOLS

Automation: PLC/HMI (Siemens STEP 7, Schneider EcoStruxure), control panels, inverters, robotic sorting (Yaskawa)

Manufacturing: CNC, laser & EDM, additive manufacturing, Autodesk Inventor, SolidWorks, AutoCAD, Rhino, ANSYS

Materials Science: Raman/IR/UV spectroscopy, DSC, TGA-DTA, XRF, XRD, AES, PVD (PLD, MBE), CVD, sol-gel

Software: MATLAB, Origin, Fortran90, LabVIEW, C#, C++, JAVA, HTML, .NET, SQL, JavaScript, Arduino

CERTIFICATIONS & MEMBERSHIPS

EUR ING (Engineers Europe) · Technical Chamber of Greece (TEE) · Materials Research Society (MRS) · IEEE

SELECTED PUBLICATIONS

- L. Kastanis et al., "Memory performance of MOS structure embedded with laser annealed gold NCs," *Solid State Electronics* 148 (2018) 63–69. DOI: 10.1016/j.sse.2018.07.012
- Doctoral dissertation: "MOS Structure with Laser Annealed Metal Nanocrystals Non-Volatile Memory," NTUA, 2021.
- Poster presentations at MRS Fall Meeting (Boston, 2015), NANOSEA (Italy, 2016), ISDRS (Maryland, 2016); oral presentations at the 32nd & 33rd Panhellenic Conferences on Solid State Physics & Materials Science.